

AVN Brake/Yaw HPP
B4137-02

Version:	02	29.06.09	DB
Description:	new		

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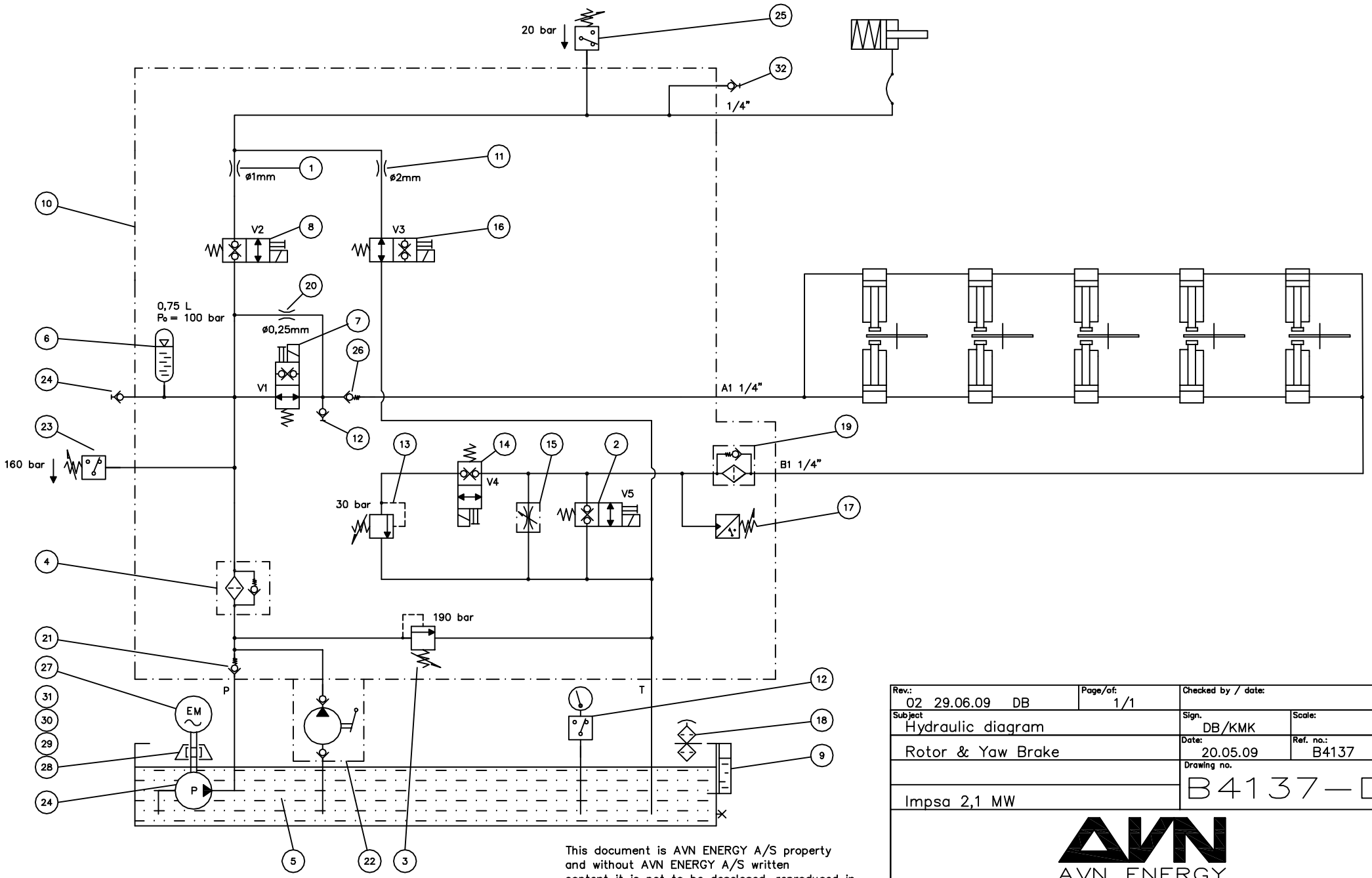
AVN Energy A/S
Suensonsvej 14
Silkeborg

AVN Energy A/S

BOM Item No.	B4137
VERSION NO.	2
DIAGRAM	
DATE	01-07-09
PAGE	1

AVN Brake/Yaw HPP Impsa 2,1 MW

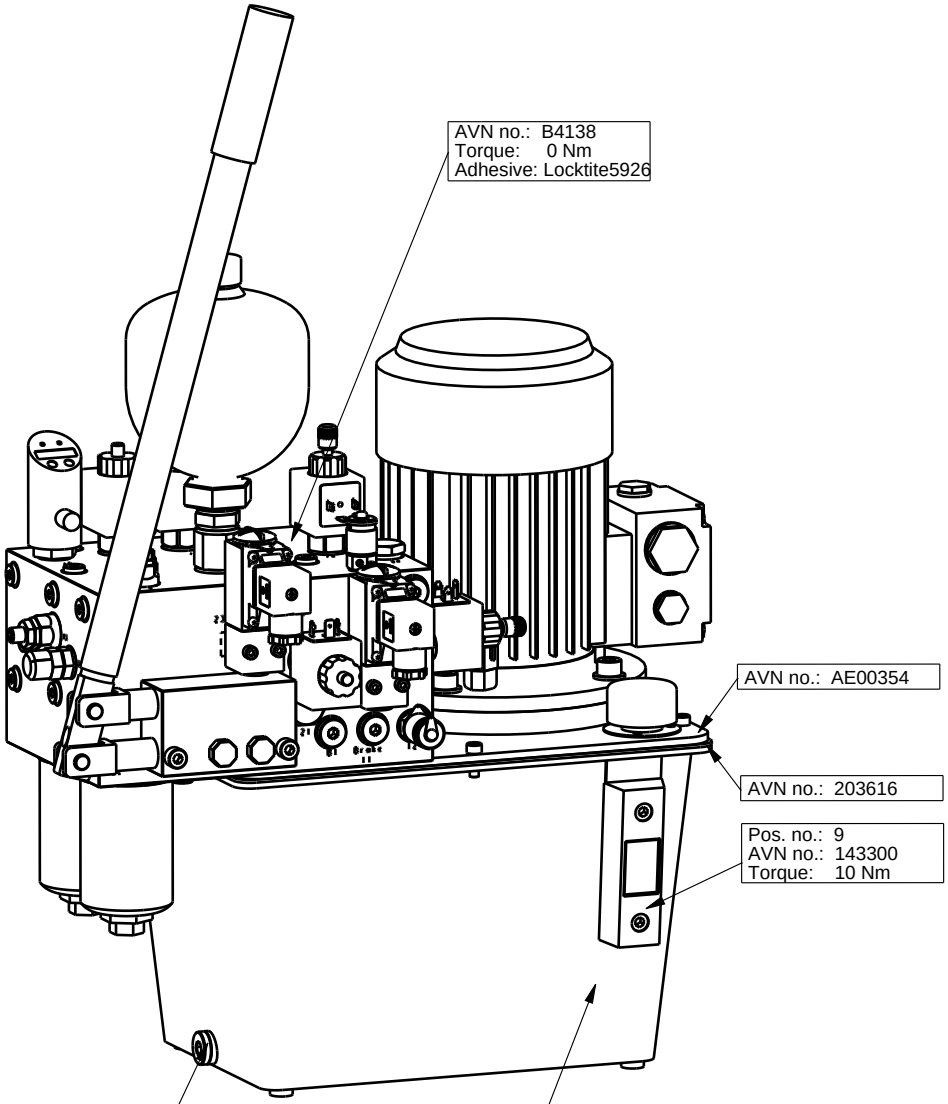
POSITION	ITEM NO.	DESCRIPTION	QUANTITY	UNIT OF MEASURE
	B4138	Subassembly for B4137	1	PCS
	8022 17 00	Socket Head Port Plug	1	PCS
	203611	MALING AF ALU TANK 11/CP10	1	PCS
	AE00354	AE Tanklid	1	PCS
	479100	GASKET RING PP45D 1/2"	1	PCS
	537120	Paint RAL9011	1	PCS
	203616	RUBBER SEAL FOR TANK NG11/CP10	1	PCS
	225220	M6 X 22 SCREW (PUMP)	4	PCS
	561702	WASHER M6 DIN125B A4-70	4	PCS
	179400	ELBOW JOINT FG 3/8-30	1	PCS
	179510	Flange for gear pump	1	PCS
	589700	SHC 524 HYDRAULIC OIL	1	L
	AH3122	AH Suction pipe	2	PCS
	360665	ORFS Male stud connector	1	PCS
	AE00041	AH Suction pipe	1	PCS
	225105	Hexagon socket screw	5	PCS
	227720	Hexagon socket screw	4	PCS
	1001035	Hexagon socket screw	4	PCS
	561712	WASHER M10 DIN 125B A4-70	4	PCS
	229060	M 10 X 25 Unb DIN 912 A4 70	4	PCS
	AE00372	AE Pressure pipe	1	PCS
	227720	Hexagon socket screw	4	PCS
	1000948	Paint RAL7005	1	PCS
	1000949	Paint RAL7005	1	PCS
5	203610	ALUMINIUM TANK NG 11/CP10 11L	1	PCS
9	143300	FBO OIL LEVEL GAUGE LVO 76/10	1	PCS
12	186522	UNS1000-MS-0122-196+temp	1	PCS
18	137700	FILLER CAPS/ BREATHER TCO 301	1	PCS
24	560002	Gear pump group 1	1	PCS
27	107703	7AA80M04-B5-2/4 0,75Kw	1	PCS
28	167922	Bellhouse Gearpump Grp.1	1	PCS
29	171100	ZK19 92shA COUPLING RUBBERSPID	1	PCS
30	171299	Coupling motor	1	PCS
31	171303	Coupling pump	1	PCS

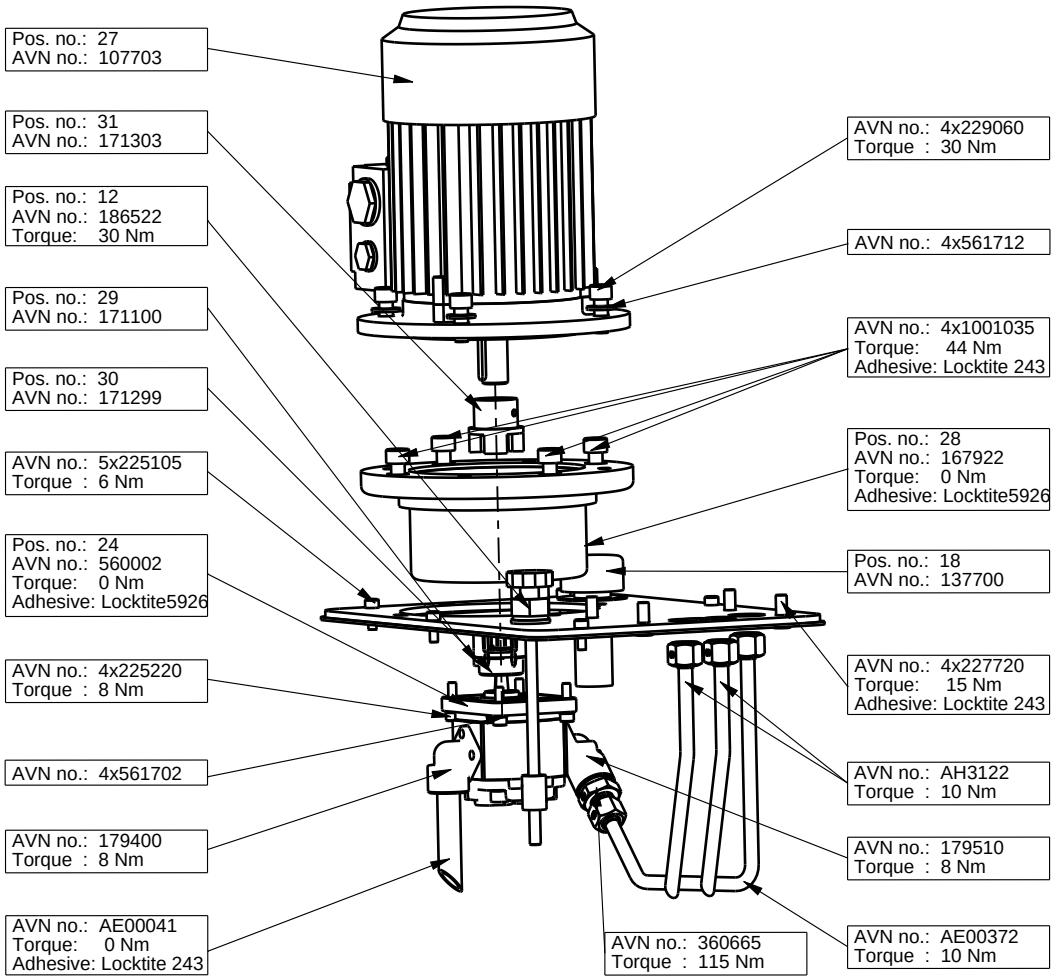


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Rev.: 02 29.06.09 DB	Page/of: 1/1	Checked by / date:
Subject: Hydraulic diagram	Sign.: DB/KMK	Scale:
Rotor & Yaw Brake	Date: 20.05.09	Ref. no.: B4137
	Drawing no.: B4137–D	
Impsa 2,1 MW		

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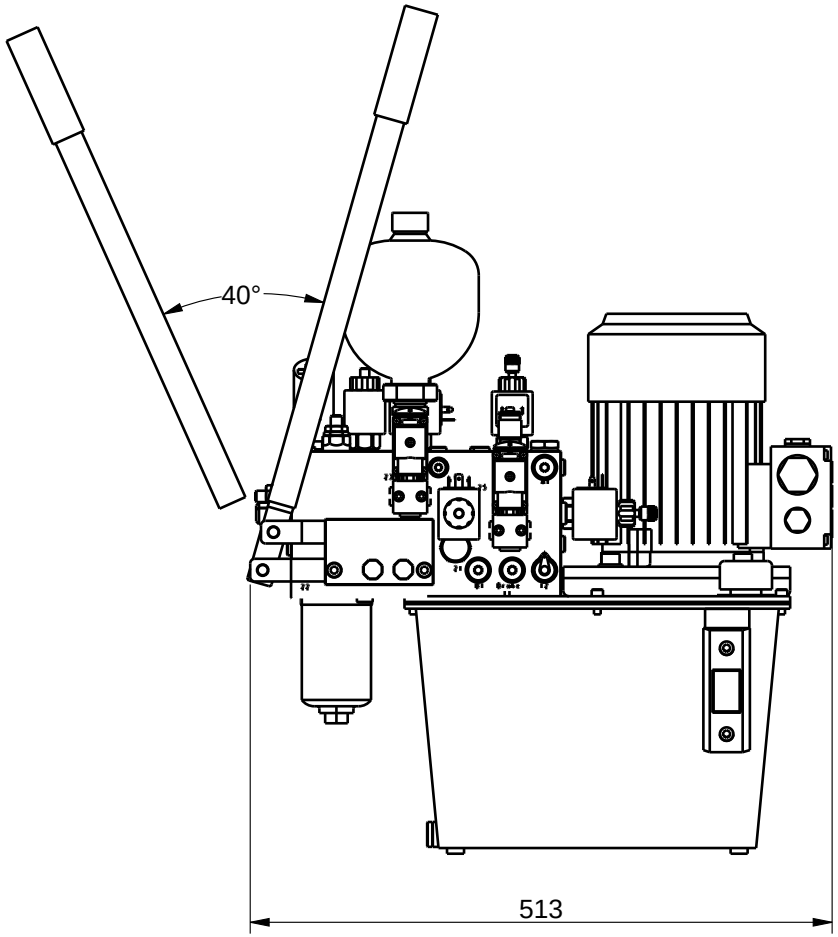


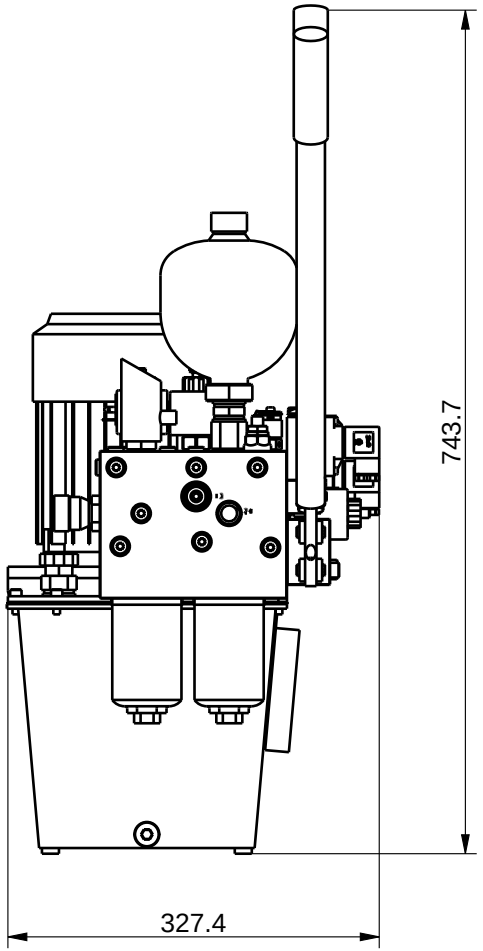
Rev.: 02 29.06.09 DB	Page / of: 1/2	Checked by / date:
Subject: Rotor & Yaw brake	Sign.: DB	Scale: 1:5
	Date: 08.06.09	Ref.no.: B4137
		Drawing no.: B4137
Impsa 2.1 MW		

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Rev.: 02 29.06.09 DB	Page / of: 2/2	Checked by / date:
Subject: Rotor & Yaw brake		Sign.: DB / Scale: 1:5
		Date: 08.06.09 / Ref.no.: B4137
		Drawing no.: B4137
Impsa 2.1 MW		

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Maintenance and replacement intervals

- Hoses must be replaced every 7. year (latest)
- AC-Motors with closed lifetime lubricated bearings must be renovated at the AVN facility after 40000 running hours (latest).
- DC-motors must be serviced after 500 running hours Brushes and commutator must be inspected and cleaned carefully, bearings and connection to pump must be inspected, if there's any doubt replaced or sent for renovation at the AVN facility.
- Accumulator precharge of nitrogen gas (N) must be measured and charged every 1/2 year.
- Gear pumps we recommend that surveillance of charging (running) if these factors start to increase significantly then replace.
- Filters must be changed every year.
- Oil quality must always be in order run test regularly e.g. every ½ year, or alternatively change the oil every 2. year.
- The oil must at all times keep a cleanliness level of 17/15/12 according to ISO 4406.
- Solenoid operated valves has a standard lifetime expectancy of 1 mill. activations must be changed before that.

IMPORTANT - IMPORTANT - IMPORTANT

It is very important when connecting additional equipment together with units delivered by AVN Energy A/S to respect following:

All hydraulic parts shall be cleaned so it at all times keeps a cleanliness level of 17/15/12 according to ISO 4406 (1999).

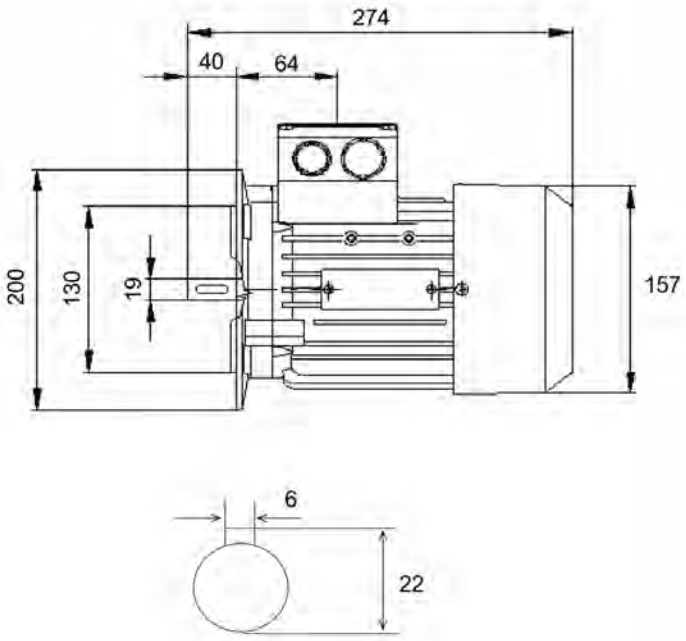
Warranty

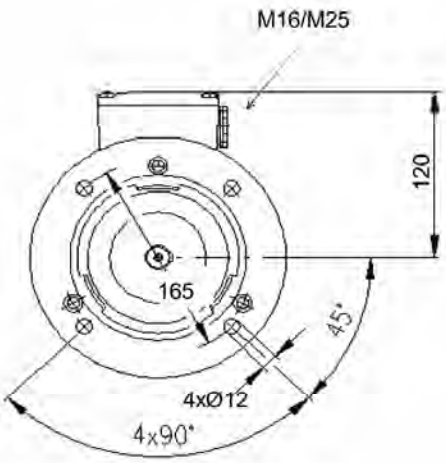
IMPORTANT: The following must be observed, if not then all warranty will fail!

- Only original AVN spare parts and consumable parts can be used in this hydraulic unit!
- All assembled subsystems must be flushed to specified cleanliness level before assembly.
- If this hydraulic unit is mal handled.
- If the operational pattern and environment differs from the original agreed.
- Only the specified hydraulic oil is to be used and the oil change intervals must be followed.

Index component data sheets:

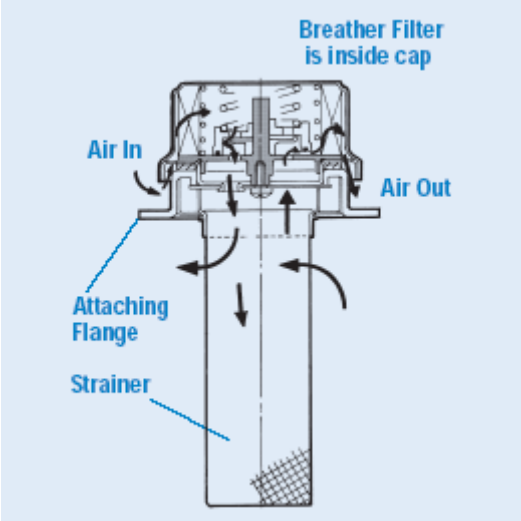
POS. :	DESCRIPTION:	PAGE/ ITEM NO.
27	7AA80M04-B5-2/4 0,75Kw	107703
18	FILLER CAPS/ BREATHER TCO 301	137700
9	FBO OIL LEVEL GAUGE LVO 76/10	143300
28	Bellhouse Gearpump Grp.1	167922
29	ZK19 92shA COUPLING RUBBERSPID	171100
30	Coupling motor	171299
31	Coupling pump	171303
12	UNS1000-MS-0122-196+temp	186522
5	ALUMINIUM TANK NG 11/CP10 11L	203610
24	Gear pump group 1	560002

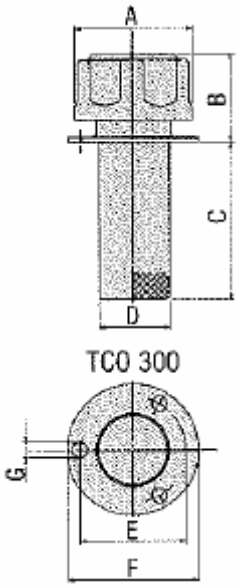




Output	0,75 Kw	**Voltage**	400 V	**Frequency**	50 Hz
Speed	1395 O/min.	**Current (In)**	1,86 A	**Mass**	9,4 Kg
Torque	5,1 Nm	**Efficiency**	72,0 %	**Duty**	S1
Noise	51,0 db(A)	**Cos f**	0,81	**Start C** **Duty C**	

	Startcurrent	**Starttorque**	**Max. torque**	**Brake**
Factor	4,2	2,3	2,3	
Calculated	7,81 A	11,7 Nm	11,7 Nm	



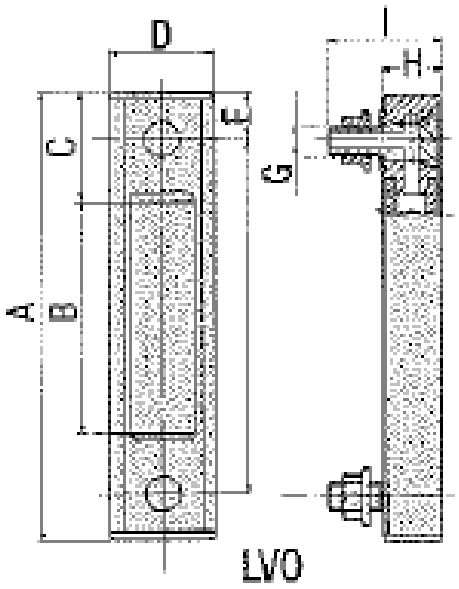


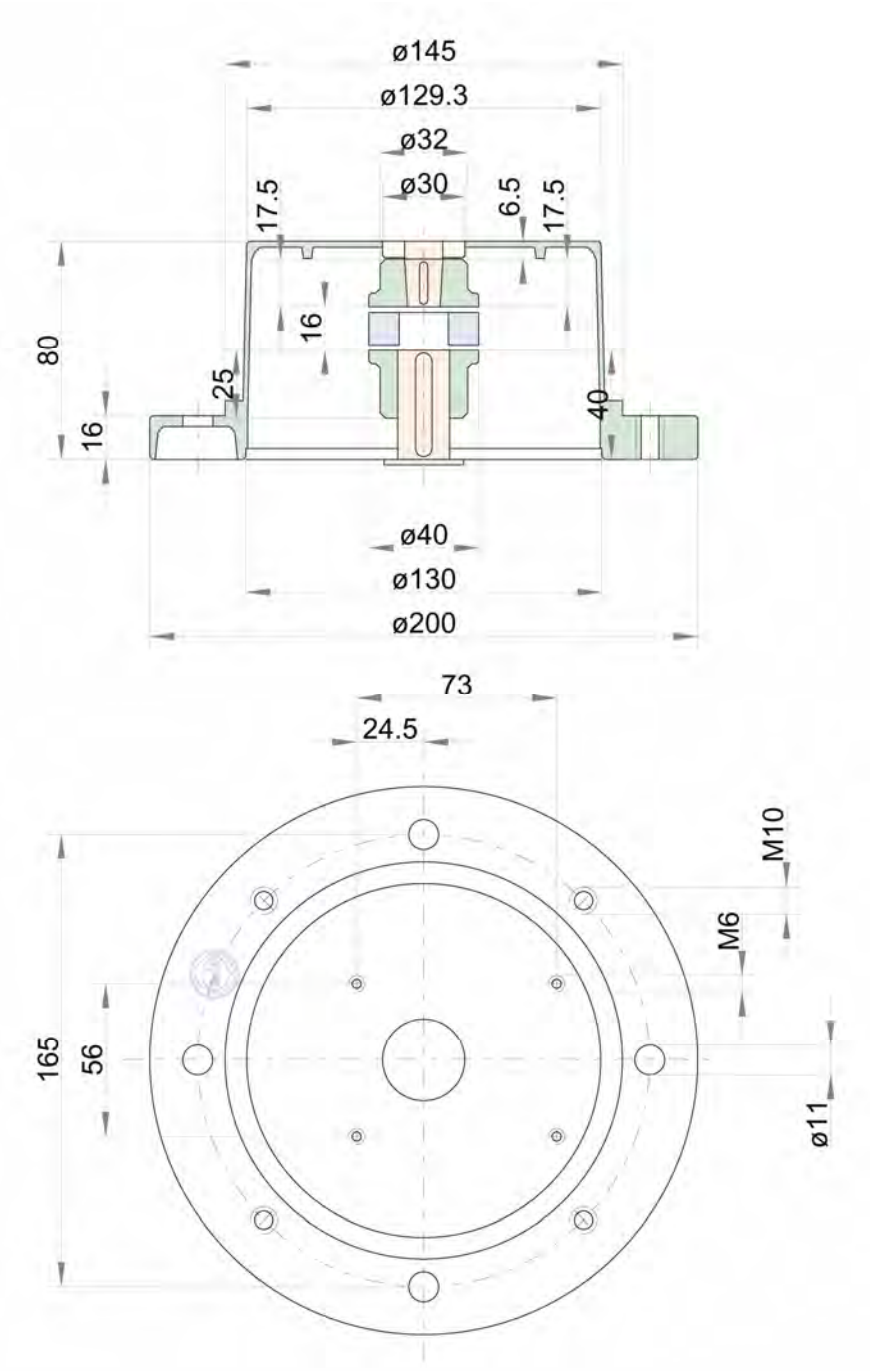
Flow [l/min]	Mesh basket* [micron]	Air filter [micron]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	F [mm]	Hole dia. [mm]
270	500	10	46	35	63	28	41	51	5,5

* Not removable

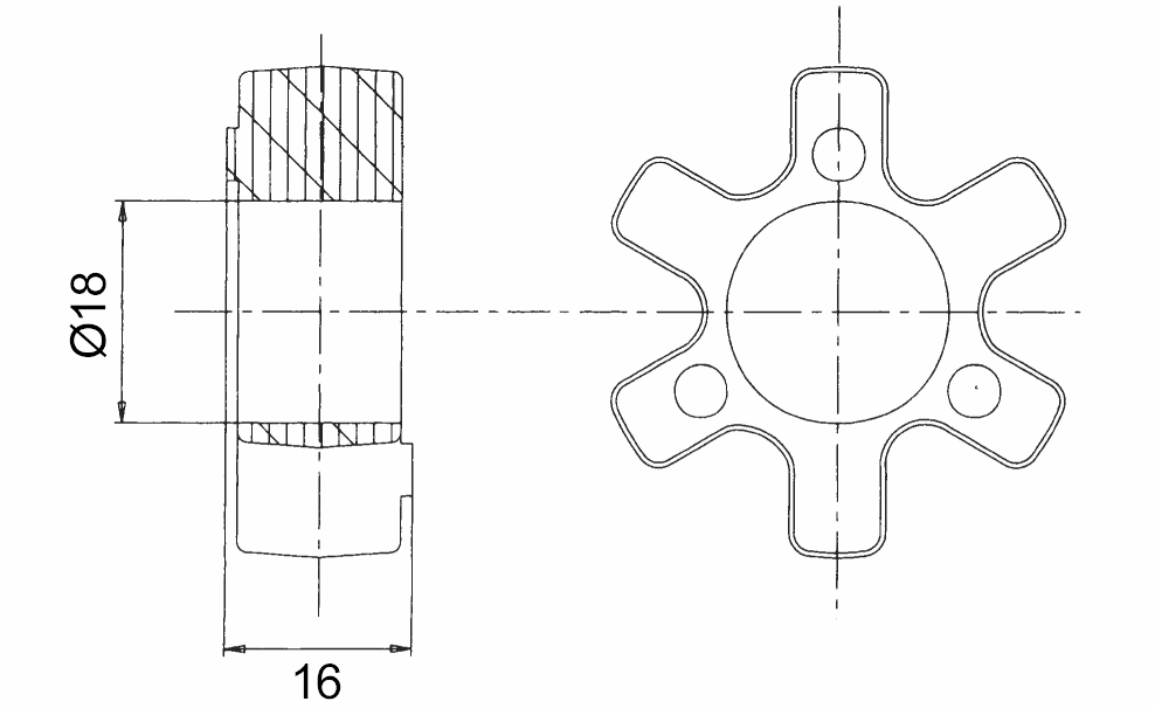
- Maximum working pressure: 100kPa / 1 bar for pressurised tanks.
- Recommmended bolt tightening torque: 10Nm, with inside nut.

	Dimensions [mm]								
Type	A	B	C	D	E	F	G	H	I
P171913 LVO 76/10	107	31	22,5	40	15,5	76	M10	20	40





Pump	MARZOCCHI 1*
IEC (50Hz)-Motor	1500 RPM 0,75 kW 80 IM B5/V1 ø200
Coupling	Spidex A19. 19H7-N1D Alu rubber spider 92° (white)

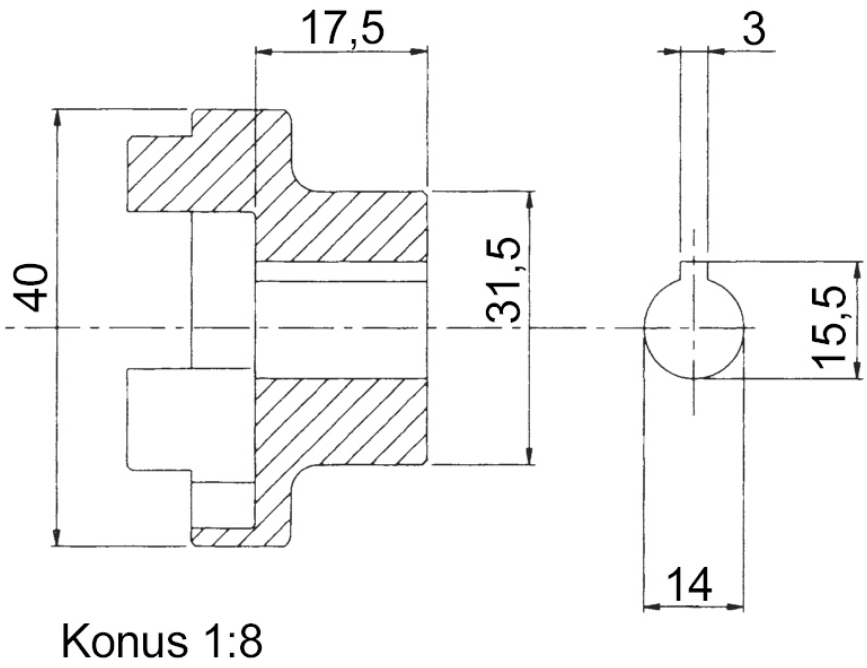


Spider 92°Shore	Torque [Nm]			Max rotation n[1/min]		Torsional angle	
	Contin. Tkn	Manimum Tkmax	Alternat. TkW	V= 30 m/s	V= 40 m/s	TkN Φkn	Tkmax Φkmax
Scale A Color: White	10	20	2,6	14000	19000	3,2°	5°

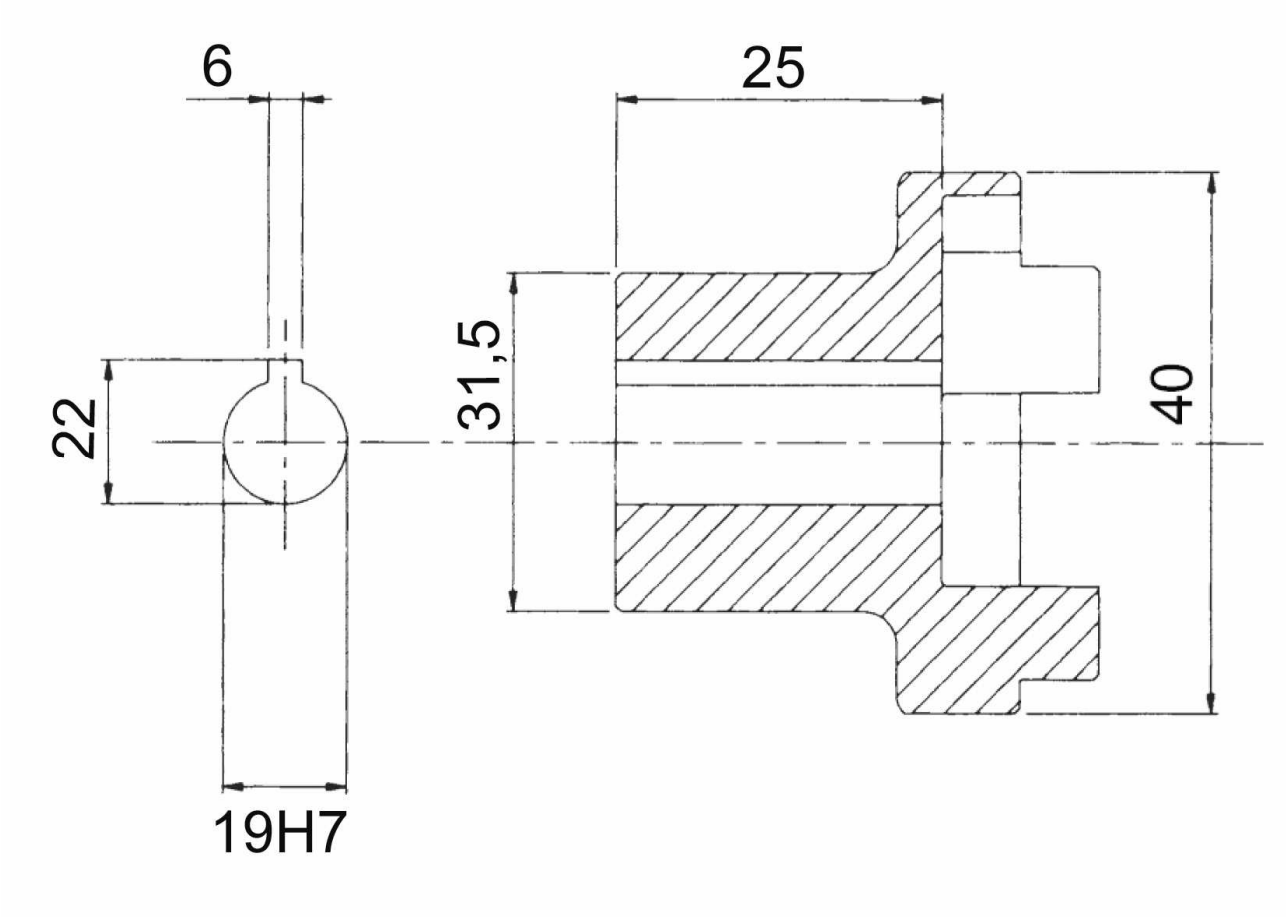
Torsional stiffness C dyn [Nm/rad]				Relatively damping
1,00 Tkn	0,75 Tkn	0,50 Tkn	0,25 Tkn	
1,28x10	1,05x10	0,80x10	0,47x10	0,75

Operation conditions. (Basic version)

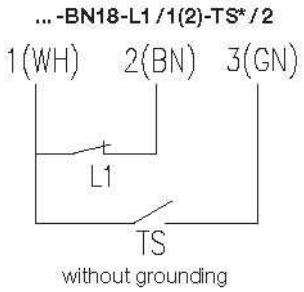
Material: .....Polyurethane
Hardnessof spider: .....92° Shore A
Permissible durable temperature range: ....-40℃ up to +90℃
Permissible short term temperature peaks:-50℃ up t o + 120℃
Damping:.....Good
Elasiticty:.....Medium
Abrasion resistance:.....Very good
Durability: .....Very good



Materiale: Aluminium



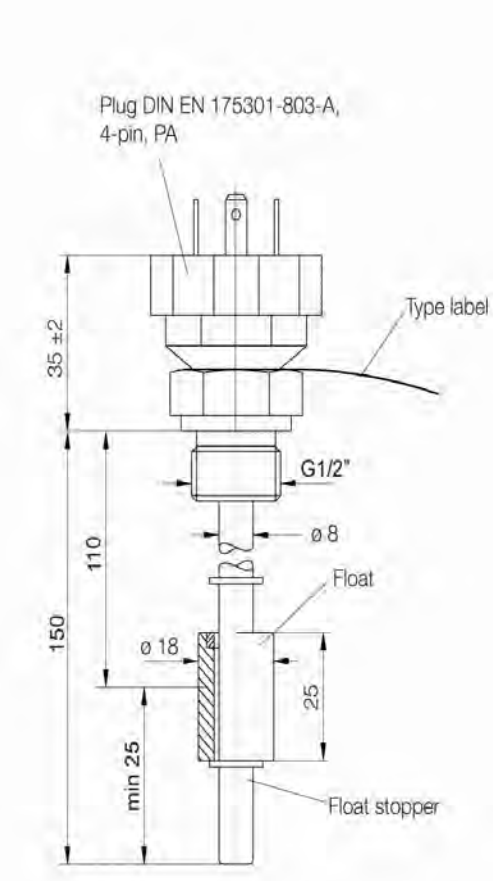
Material: Aluminium





Technical Data

Materials Bulkhead union: Contact tube: Float stopper: Float:	 Brass Brass Bronze Foamed NBR
System of protection	IP65
Housing	——
Process connection Male thread:	 G1/2”
Electrical connection	Plug DIN EN 175301-803-A (former DIN 43650), 4-pin, PA
Electrical valves	max 24 V AC/DC max 1 A max 20 VA/W
Operating temperature	-10℃ ... +90℃
Operating pressure	max 4 bar (400 kPa = 58 psi)
Density	min 0,64 g/cm3
Depth of immersion at density 1	15mm ±2
Temperature switch **Temperature switch :** **Indexing tolerance :** **Differential gap :** **Contact :**	Bimetal +70°C ± 5K 30 ± 15K NC (the contact is closed at room temp)



Completes oil tanks type CP10GC

Made of die-cast aluminium. Supplied complete of CCP10, cover in steel plate, GCP10 gasket in "guarnital". Nominal capacity of 10 dm³, corresponds to 3/4 of total capacity.

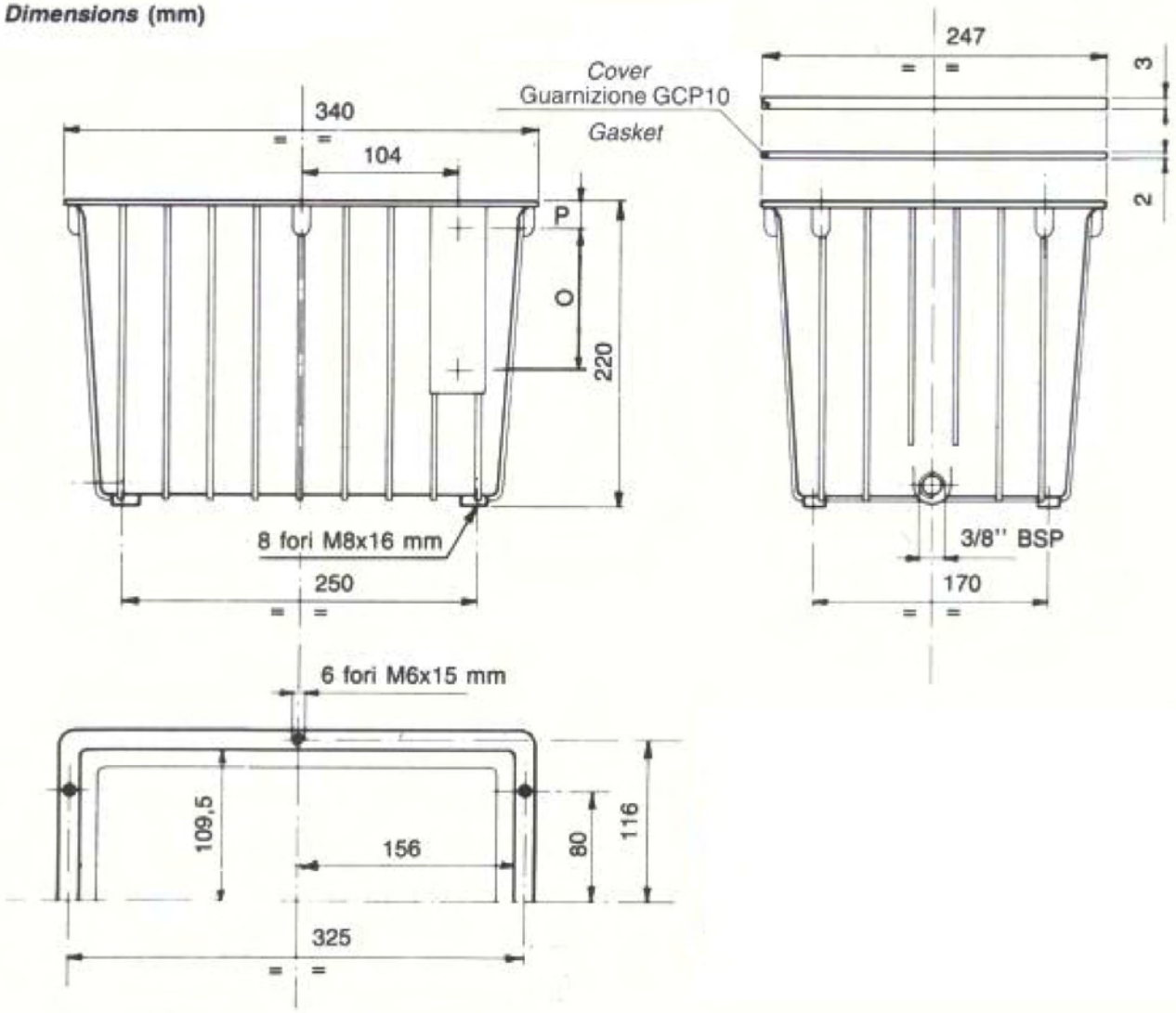
Oil tanks type CP10

Made of die-cast aluminium. They are supplied with drain plug.

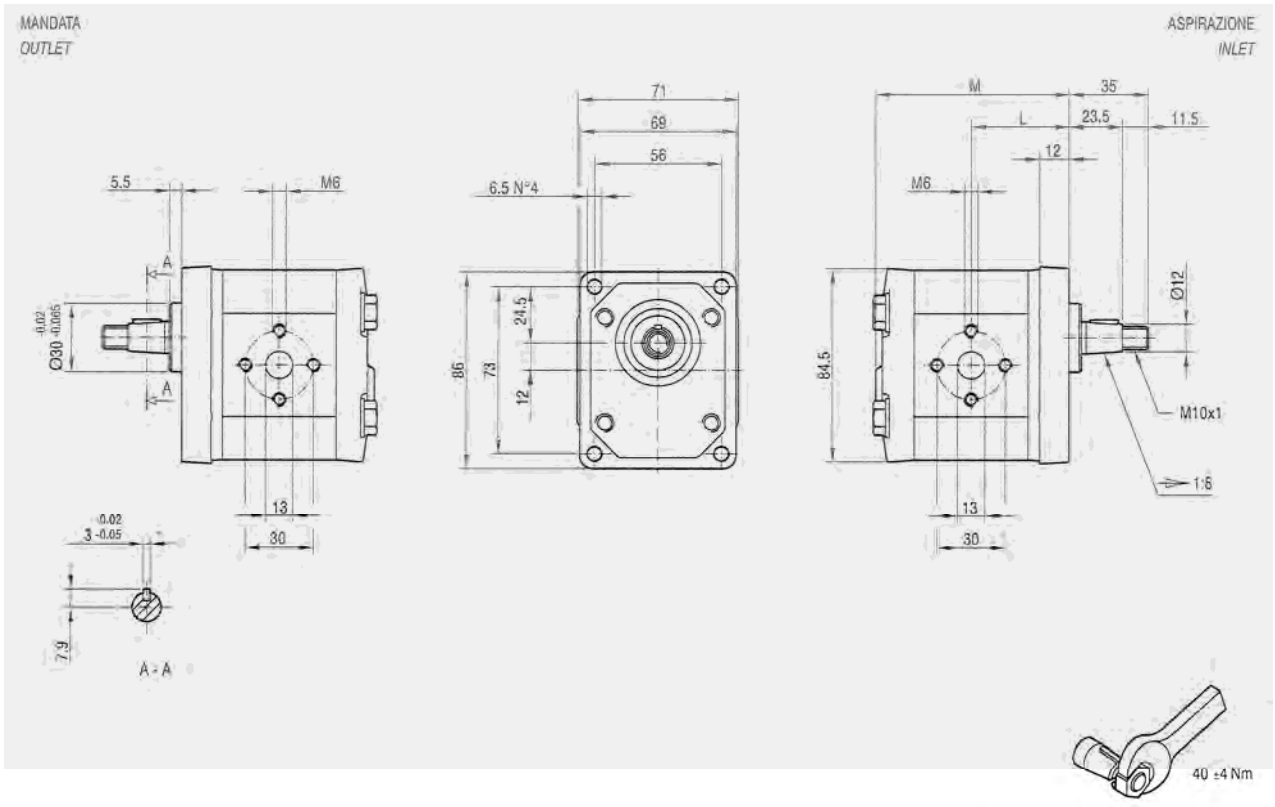
Gasket type GCP10

Made of 2 mm thick "guarnital" material

Dimensions (mm)



Gearpump
- Clockwise rotation
- European flange
- 1:8 tapered shaft, flanged port, type E
- Standard seals
- Die-cast steel flange and cover



CILINDRATA *DISPLACEMENT*	PORTATA a 1500 giri/min *FLOW at 1500 rev/min*	PRESSIONI MASSIME *MAX PRESSURE*			VELOCITÀ MASSIMA *MAX SPEED*	DIMENSIONI *DIMENSIONS*	
		P_1	P_2	P_3		L	M
cm^3/giro [cm^3/rev]	litri/min [litres/min]	bar	bar	bar	giri/min [rpm]	mm	mm
1,4	2,0	270	290	310	6000	40	79